charges

Charge is a quantity that is **conserved** and **quantised** (discrete). This means that you **cannot** have a fraction of a charge; $\text{charge} \in \{0, \mathbb{Z}^+\}$.

Opposite charges attract; like charges repel.

For a positive test charge between one positive charge and one negative charge (as an example), the positive test charge will be **repelled** by the **positive charge** and **attracted** by the **negative charge**. This means that the net field strength and **net force** acting on the test charge are obtained from **vector sums**, respectively (and independently); for either value, there would be two parts: amount caused by the positive charge and the amount caused by the negative charge.

Charge can be transferred through:

- **Friction**
- **Contact**; one charged object is touched against an uncharged object. Electrons are transferred to, or from, the uncharged object.
- **Electrostatic induction**; a charged object induces a charge in a **neutral object without touching it**.
  *Remember: Fried **C**hicken **E**nergy*

When describing the movement of charge in conductors, remember that it is always the electrons that move. Positive charges in materials are generally attached to the atoms. Positive charge is the **absence of electrons**.

In the process of **electrostatic induction**, the conductor is connected momentarily to Earth with a **conducting wire**. This allows electrons to flow to, or from, the ground (depending on the charge of the rod) until the **system** has no further ability to move charge.

The process of connecting a conductor to Earth is called **earthing** or **grounding**.
Earth is regarded as an infinite source, or sink, of electrons.

The law that describes the variation with distance of the force between two point charges is known as Coulomb's law.

When charges move in current electricity, energy must be transferred to them from the source of emf in the circuit. Hence, there exists energy that is gained by individual electrons, albeit a very small amount because the work done on the electron is eV.

One electron volt (abbreviated eV) is the energy transferred by one electron as it moves through a potential difference of one volt.

In atomic and nuclear physics, it is usual to replace the joule as the unit of energy with the electronvolt, where $1 \text{ eV} = 1.60 \times 10^{-10} \text{ J}$.

Moving charged particles are affected by the presence of **electric and magnetic fields**. The fields lead to forces that act on the particles.